Ocular Trauma

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Ocular trauma = common emergency topic! Must know first aid steps!

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Classification — Blunt vs Penetrating

BETT classification is
the standard system!
Open vs closed globe!

Birmingham Eye Trauma Terminology (BETT)

Category	Type	Definition
Closed globe	Contusion	No full-thickness wound — blunt force
Closed globe	Lamellar laceration	Partial-thickness wound of eye wall
Open globe	Rupture	Full-thickness wound by blunt object
Open globe	Penetrating	Single entry wound (no exit wound)
Open globe	Perforating	Entry + exit wound (through and through)
Open globe	IOFB	Retained foreign body inside the eye

Effects of Blunt Trauma (Anterior to Posterior)

Structure	Injury
Lids	Ecchymosis (black eye), laceration, emphysema
Cornea	Abrasion, edema, Descemet tears (Haab striae in buphthalmos)
Anterior chamber	Hyphema, angle recession, iridodialysis
Iris	Iridodialysis (tear from root), traumatic mydriasis
Lens	Subluxation, dislocation, rosette cataract, Vossius ring
Vitreous	Hemorrhage
Retina	Berlin edema (commotio retinae), retinal tears/detachment
Choroid	Choroidal rupture (crescent-shaped, concentric to disc)
Optic nerve	Traumatic optic neuropathy

Hyphema = blood in AC!
Grading + management
= exam favorite!

Hyphema — Grading System

Grade	Blood Level	Management
Microhyphema	RBCs in AC only (no level)	Conservative — rest, cycloplegic, steroid
Grade I	< 1/3 of AC filled	Conservative — bed rest, head elevation 30-45
Grade II	1/3 to 1/2 of AC	Conservative + close monitoring of IOP
Grade III	> 1/2 but not total	May need AC washout if IOP uncontrolled
Grade IV	Total (8-ball hyphema)	AC washout — risk of corneal blood staining

Complications of Hyphema

- Re-bleed (secondary hemorrhage): 3-5 days, more severe than primary
- Secondary glaucoma: blood blocks trabecular meshwork
- Corneal blood staining: prolonged hyphema + raised IOP
- Optic atrophy: from sustained raised IOP

Berlin Edema (Commotio Retinae)

Retinal whitening after blunt trauma, especially at posterior pole (macula).

Caused by disruption of photoreceptor outer segments. Usually self-limiting.

If macula involved -> permanent central vision loss possible (cherry red spot appearance).

Angle Recession Glaucoma

★ **Tear between longitudinal and circular fibers of ciliary muscle after blunt trauma.**

Causes late-onset open angle glaucoma — may present months to YEARS after trauma.

Diagnosed by gonioscopy: widened ciliary body band. Lifelong IOP monitoring needed!

Chemical burns = TRUE EMERGENCY! Irrigate first ask questions later!

Alkali vs Acid Burns

Feature	Alkali Burns	Acid Burns
Agents	NaOH, Ca(OH)2, NH3	H2SO4, HCl, HF
Mechanism	Liquefactive necrosis	Coagulative necrosis
Penetration	Deep (saponifies fats)	Superficial (protein barrier)
Severity	MORE dangerous	Less dangerous (except HF)
Prognosis	Worse — ongoing damage	Better — self-limiting

★ **FIRST AID: Immediate copious irrigation with water/saline for 30+ minutes!**
Check pH every 15 min until neutral (7.0-7.4). Do NOT wait to find specific antidote.

Roper-Hall Classification of Chemical Burns

Grade	Cornea	Limbal Ischemia	Prognosis
I	Clear cornea	No ischemia	Good
II	Hazy, iris visible	< 1/3 (< 4 clock hrs)	Good
III	Hazy, iris obscured	1/3 to 1/2 (4-6 clock hrs)	Guarded
IV	Opaque cornea	> 1/2 (> 6 clock hrs)	Poor

Management of Chemical Burns (After Irrigation)

Drug / Procedure	Purpose
Topical antibiotics	Prevent secondary infection
Cycloplegics (atropine)	Pain relief + prevent synechiae
Topical steroids (1st week)	Reduce inflammation (taper after 10 days!)
Vit C (topical + oral)	Promotes collagen synthesis for corneal healing
Citric acid 10%	Chelates calcium in alkali burns
Amniotic membrane graft	For non-healing epithelial defects
Limbal stem cell transplant	For severe limbal ischemia (Grade III-IV)

Metallic IOFB = chronic toxicity to eye tissues!
Siderosis vs Chalcosis!

Intraocular Foreign Body (IOFB)

Feature	Detail
Common setting	Hammering, chiseling, industrial work, blast injury
Diagnosis	X-ray orbit, CT scan (gold standard), USG B-scan
MRI	CONTRAINDICATED if metallic FB suspected!
Entry sign	Small corneal/scleral wound, shallow AC, lens opacity
Management	Remove IOFB by electromagnet (anterior) or vitrectomy (posterior)

Siderosis vs Chalcosis — The Comparison

Feature	Siderosis (Iron)	Chalcosis (Copper)
Metal	Iron (ferrous)	Copper (cuprous/cupric)
Mechanism	Iron deposited in tissues	Copper deposited in tissues
Iris	Heterochromia (rust-brown)	Green discoloration
Lens	Brown discoloration	Sunflower cataract
Retina	Pigmentary degeneration	Kayser-Fleischer-like ring
ERG	Extinguished (b-wave gone)	May be normal early
Prognosis	Progressive — remove FB	Depends on purity of copper
Association	Industrial trauma	Wilson disease (endogenous copper)

★ **Pure copper (>85%): causes fulminant endophthalmitis — needs emergency removal!**
Copper alloy (<85%): causes chalcosis (chronic, progressive) — still needs removal.
Gold, silver, platinum, glass: inert — may be left in eye if asymptomatic.

Trauma = emergency topic!
Know first aid cold!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Classify ocular injuries (BETT). Describe management of chemical burns
- 2. Describe chemical burns of eye — Roper-Hall classification & management
- 3. Describe effects of blunt trauma on anterior segment. Discuss hyphema
- 4. Write about intraocular foreign body — diagnosis, siderosis, management

Most Repeated SAQs (5–6 marks)

- 1. Hyphema — grading and management
- 2. Siderosis vs Chalcosis (table format)
- 3. Roper-Hall classification of chemical burns
- 4. Berlin edema (commotio retinae)
- 5. Angle recession glaucoma
- 6. Sympathetic ophthalmia (cross-topic with uveitis)

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
First aid in chemical burns	Immediate copious water irrigation
Alkali burns worse because	Liquefactive necrosis (deep penetration)
Rosette cataract	Blunt trauma (concussion)
Vossius ring	Iris pigment imprint on lens (blunt trauma)
8-ball hyphema	Grade IV (total AC filled with blood)
Sunflower cataract	Chalcosis (copper) / Wilson disease
Siderosis — diagnostic test	ERG (extinguished b-wave)
MRI contraindicated in	Metallic IOFB (iron/steel)
Berlin edema location	Posterior pole (retina)
Angle recession — late complication	Open angle glaucoma (years later)
Steroid use after chemical burn	Only first 7-10 days (risk of melting)

♥ Quick first aid saves eyes, Sammy! You'll be the calm doctor in the ER who knows exactly what to do. That confidence starts here!